A proof of the conjectured recurrence for OEIS A001712

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25 August 2026

Abstract

OEIS A001712 carries a conjectured linear recurrence with polynomial coefficients, of order 3, contributed by its contributor. We prove it. The generating function of the entry is algebraic, and the recurrence is equivalent to the vanishing of an explicit differential operator applied to it: writing $\theta = x\,\frac{d}{dx}$, the sequence $b(n) = \sum_i p_i(n)a(n-i)$ has generating function $B(x) = \sum_i x^i\,(p_i(\theta+i)A)(x)$, so the conjecture holds for all $n > d$ exactly when B is a polynomial of degree d. Since A lies in a quadratic extension of $\mathbb{Q}(x)$, which is closed under θ, the residual B can be computed in closed form. Here it equals 0, a polynomial of degree 0, which proves the recurrence for every $n > 0$.

2020 Mathematics Subject Classification. 11B37, 05A15, 33F10.

1 The sequence and the conjecture

OEIS A001712 is "Generalized Stirling numbers, [n+5,5]_3". It has offset 0 and begins

$$a(0), \ldots, a(7) \;=\; 1, 12, 119, 1175, 12154, 133938, 1580508, 19978308, \;\ldots$$

The entry gives the exponential generating function

$$A(x) \;=\; \sum_{n\geq 0} a(n)\,\frac{x^n}{n!} \;=\; \frac{-6\log{(1-x)}^2 + 7\log{(1-x)} - 1}{(x-1)^5}, \tag{1}$$

and it carries the following comment:

> **Conjecture:** a(n) + 3*(-n-3)*a(n-1) + (3*n^2 + 15*n + 19)*a(n-2) - (n+2)^3*a(n-3)=0.
> - *R. J. Mathar*, Jun 09 2018

As of the "Last modified" line on the live entry (, revision 0) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^{3} p_i(n)\,a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = 1, \qquad p_1(n) = -3\,(n+3)\,, \qquad p_2(n) = 3n^2 + 15n + 19, \qquad p_3(n) = -\,(n+2)^3$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n\geq 0} a(n)x^n/n!$ be the exponential generating function of a sequence, and let $\theta = x\,\dfrac{d}{dx}$, so that θ acts on x^m as multiplication by m. For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m)a(m)x^m/m!$.

Lemma 1. *Let $p_0, \ldots, p_r$ be polynomials and suppose the conjecture asserts $\sum_{i=0}^{r} p_i(n)a(n-i) = 0$. Re-index by $n = m + r$ and $j = r - i$, and write $q_j(m) = p_{r-j}(m + r)$, so the claim reads $\sum_{j=0}^{r} q_j(m)\, a(m + j) = 0$. Put $b(m) = \sum_j q_j(m)a(m + j)$. Then*

$$B(x) := \sum_{m\geq 0} b(m)\frac{x^m}{m!} = \sum_{j=0}^{r} \big(q_j(\theta)A^{(j)}\big)(x),$$

where $A^{(j)}$ denotes the j-th derivative of A.

Proof. For an exponential generating function an upward shift is differentiation: $\sum_{m\geq 0} a(m + j)x^m/m! = A^{(j)}(x)$. Multiplying the coefficient of $x^m/m!$ by $q_j(m)$ is exactly the action of $q_j(\theta)$, and summing over j gives the statement. The re-indexing is what makes this work: a downward shift $a(n-i)$ would correspond to repeated integration, whereas an upward shift is a derivative, and the field used below is closed under differentiation. □

Corollary 2. *With the notation of Lemma 1, the recurrence $\sum_i p_i(n)a(n - i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d.*

Proof. $b(m)$ is $m!$ times the coefficient of x^m in B, so $b(m) = 0$ for all $m > d$ says exactly that B has no terms above x^d. □

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A, and one only has to recognise it.

Lemma 3. *Let $D \in \mathbb{Q}(x)$ and let $K = \mathbb{Q}(x)\big[\sqrt{D}\big]$. Then K is closed under θ and under d/dx; explicitly,*

$$\theta\big(u + v\sqrt{D}\big) = x\, u' + \Big(x\, v' + \frac{x\, v\, D'}{2D}\Big)\sqrt{D}, \qquad u, v \in \mathbb{Q}(x).$$

Proof. Differentiate: $(\sqrt{D})' = D'/(2\sqrt{D}) = \big(D'/(2D)\big)\sqrt{D}$, and multiply by x. Both components remain in $\mathbb{Q}(x)$. □

Consequently, if $A \in K$ then every $p_i(\theta + i)A$ is in K, and so is B. Writing $B = P + Q\sqrt{D}$ with $P, Q \in \mathbb{Q}(x)$, the test of Corollary 2 becomes: B is a polynomial precisely when $Q = 0$ and P has constant denominator. Both are decided by exact cancellation of rational functions.

3 The computation for A001712

The generating function (1) lies in $K = \mathbb{Q}(x)[\sqrt{D}]$ with

$$D \;=\; 1.$$

Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, the $\sqrt{D}$ component of B cancels identically and the rational part collapses to

$$B(x) \;=\; 0,$$

a polynomial of degree 0.

Theorem 4. *The conjectured recurrence*

$$1\, a(n) - 3\,(n + 3)\; a(n - 1) + (3n^2 + 15n + 19)\, a(n - 2) - (n + 2)^3\; a(n - 3) \;=\; 0$$

holds for every $n > 0$.

Proof. Immediate from Corollary 2 and the displayed value of B. □

Remark 5. The polynomial B is not an error term: it records the boundary of the recurrence. For $n \leq 0$ some of the shifted terms $a(n-i)$ fall outside the range of the sequence, and B collects exactly those contributions. Above that range the relation is exact.

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, the generating function was checked against the entry rather than assumed: the Taylor coefficients of (1) were expanded and compared term by term with the DATA section of A001712, agreeing on all 13 terms compared.

Second, the recurrence itself was evaluated directly on the published terms, in exact integer arithmetic and without reference to the generating function: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 16 values of n from $n = 3$ upward. This is what Corollary 2 predicts from the degree of B, and it is a genuinely different computation from the symbolic one: it never forms B, never differentiates, and uses only integers.

The symbolic step is exact throughout. The residual is obtained by rational-function cancellation in K, not by series truncation, so the identity $B = 0$ is an identity of functions and the theorem holds for all $n > 0$ simultaneously.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A001712.

[2] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011, Chapter 7 (holonomic closure properties and the translation between recurrences and differential equations).

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999, Chapter 6 (algebraic and D-finite generating functions).